Beyond WEIRD humans and STRANGE dogs: Using big team science to improve generalizability and reproducibility in comparative psychology

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Abstract

Canine science aims to understand dogs as a species uniquely adapted to live alongside humans. Research in the field has increased rapidly in the 21st century, but has struggled with representativeness and generalizability. Here we discuss key issues and identify solutions through big team science innovation and collaboration with researchers in the Global South. Sampled populations are usually from the Global North, where researchers and guardians can be characterized by WEIRD attributes and dogs may be shaped by STRANGE factors, severely limiting generalizability across locations and the overall replicability of the science. Big team science provides an ideal avenue to overcome some of these biases and include diverse perspectives and populations, fostering global collaboration.

Shaped by millennia of natural and artificial selection pressures, dogs (*Canis familiaris*) are house pets, working companions, and free-living individuals adapted to share the human ecological niche. In research, dogs have transitioned from models of mammalian evolution (Darwin, 1872) and human learning and behavior (Seligman et al., 1968) to conscious and sentient subjects of comparative psychology that provide unique insights into cognitive evolution and ontogeny (Arden et al., 2020; Aria et al., 2021). Emerging in the 1990s, canine science highlights the importance of exploring how diverse dog populations impact local ecosystems, as well as deepens our knowledge about dog behavior and socioecology across the globe. Despite the universal need to understand factors shaping dog behavior, cognition, and welfare, canine research is largely based on Western perspectives and scientific traditions, viewing the dog-human experience through a particular cultural lens.

This narrow focus on pet dogs from the Global North, paralleling the WEIRD (**W**estern, **E**ducated, **I**ndustrialized, **R**ich, and **D**emocratic) contexts that undermine human psychological study (Henrich et al., 2010), reduces the reproducibility and broader generalizability of canine research. The analogous STRANGE framework (**S**ocial background; **T**rappability and self-selection; **R**earing history; **A**cclimation and habituation; **N**atural changes in responsiveness; **G**enetic make-up; and **E**xperience; Webster & Rutz, 2020) is especially relevant for contextualizing companion dogs as a non-representative population with multiple factors limiting inferences in comparative psychology. Therefore, while we've gained insights to the cognitive and affective experiences of humans' longest cooperative partner, this information only applies to a subgroup of the world's domestic dogs (Gompper, 2014). To move the field forward and address pressing global challenges in human-animal interaction, we need to critically evaluate the populations of dogs and guardians being sampled, address the challenges of conducting inclusive, equitable research across the Global North *and* South, and co-create innovative approaches for achieving ethical, internationally relevant research. Here, we discuss core issues limiting the replicability of canine science and the generalizability and representativeness of dogs and humans. We see a possible solution through a big team science approach such as the ManyDogs Project (ManyDogs Project, 2023a), which can advance our understanding of dog behavior and promote inclusive international collaborations, leading to a more nuanced understanding of human-animal interactions.

Generalizability Across Locations

Dogs inhabit every continent and almost every island on Earth (Wandeler et al. 1993), making them an accessible species to study worldwide. However, is a dog in New York City the same as one in Vienna, Kyoto, Kolkata, or Buenos Aires? Across the world, dogs vary substantially, making findings in one location less generalizable to others (Figure 1).

One of the most obvious and potentially strongest sources of variation in dogs is the hundreds of distinct breeds. Breeds differ dramatically not only in morphology (a 75-fold difference between smallest and largest breeds) but also in behavior. While some breeds excel at detecting scents, others show higher levels of trainability, impulsivity, or problem behaviors (Pongrácz & Dobos, 2025). Variation in breeds alone probably makes dogs the most behaviorally diverse species on the planet. And breeds are not evenly distributed across the globe, leading to variation in breed composition and hybridization in different locations. Interestingly, countries may vary in the frequency of hybrids and mixed breeds with potentially more in the Global South and the United States than Europe.

Figure 1: Dogs are found across the world with marked differences between the Global North and South in dog-human relationships, breeds and morphology, and research traditions.

Alongside pronounced dog breed differences, human guardians exhibit clear cultural differences within and between countries. The dog-human bond is a critical component of companion dogs' behavior, with guardians shaping their lives from nutrition to socialization to training. There are massive differences between and within cultures in how dogs are perceived, valued, and treated (Serpell, 2004). Some herding and guarding dogs live outside with the livestock that they tend, while some urban "fur babies" live in luxury, with every need met. The intricate layering of dog and guardian diversity results in a level of variation not observed in other species.

This extreme variation observed between rural/urban dogs and across countries and cultures is rarely accounted for in dog behavior and cognition studies. Dozens of canine science research sites exist in North America, South America, Europe, Asia, and Oceania. Would we expect behavior in one site to generalize to another? Research teams in different regions likely recruit different subpopulations of guardians. For instance, the ManyDogs 1 project recruited guardians from 20 different research sites in North America, South America, and Europe (ManyDogs Project et al. 2023b). Yet, the guardian populations differed across sites in terms of their age distribution (Figure 2A). Though New England sites had fairly uniform age distributions, New York, Budapest, and Messina skewed toward younger guardians, while Arizona skewed toward older guardians. Guardian age likely affects their dog's behavior (e.g., older, retired guardians may spend more time with their dogs) and even which breeds they selected (e.g., older guardians may choose smaller, less energetic breeds). Thus, different subpopulations of guardians can result in different subpopulations of dogs being tested across research sites.

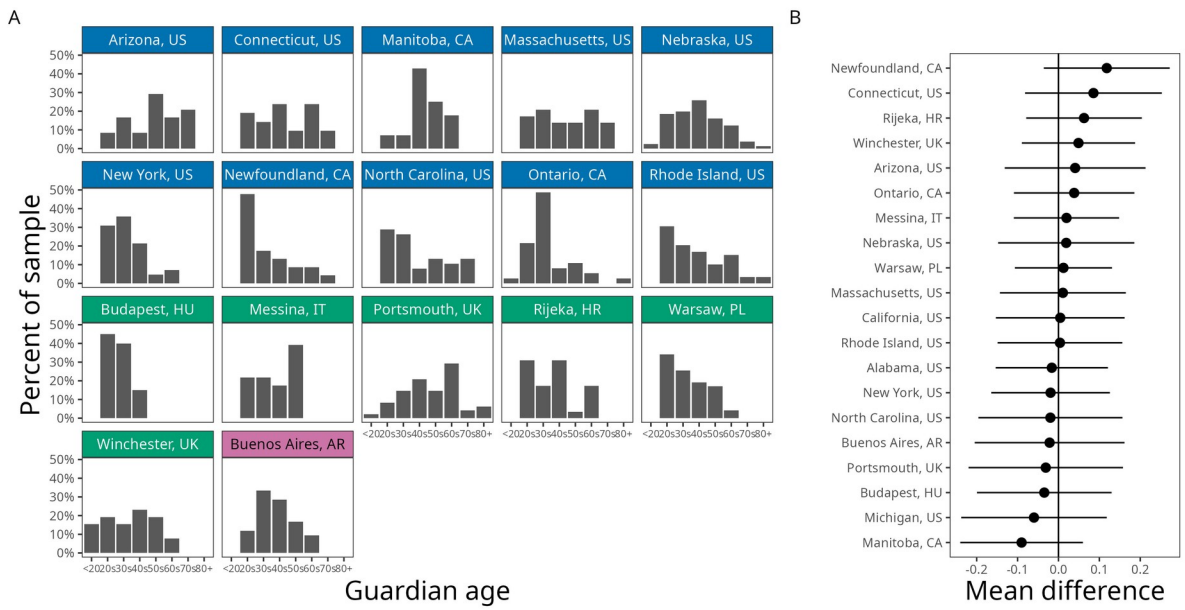

Figure 2: (A) Guardian age distribution across research sites. For the ManyDogs 1 project, 20 research sites tested pet dogs (three sites tested working dogs without owners and are not shown here). Bars represent the relative frequency of different age groups tested at each site. Sites labeled in blue are North American, green are European, and pink are South American. (B) Study effects across research sites. Dots represent mean differences between conditions. Bars represent standard deviation. In general, the variation within a site (standard deviation) is larger than the variance between sites (difference between smallest and largest mean difference).

These research site differences are important to account for because some dog behavior studies have failed to replicate (Stevens et al., 2022). Though it is possible that unreplicated effects result from idiosyncratic designs or analysis, truly absent effects, or random error, another possibility is that effects found in one sample of dogs are indicators of socioecological differences that do not generalize to all dogs. That is, research sites may be testing different subpopulations and getting different results, which could be due to breed baserates in the population, different recruitment strategies of the researchers, or broader environmental and cultural factors such as demographic differences in the human populations. However, it remains unknown whether these population differences across sites translate to differences in behavior. A big team science group studying infants found that variation within research sites exceeded variation between sites (ManyBabies Consortium, 2020), which seems to also be the case for ManyDogs 1 (Figure 2B), despite the site differences in demographics.

Representativeness in WEIRD Contexts and STRANGE Samples

Similar to other fields, canine science has inherent sampling biases influencing which guardians contribute to research (e.g., Elston, 2021; Kaźmierczak et al., 2023). This seems partially caused by location. Canine research is often conducted at university campuses in the Global North, which have proximity biases for the nearby student population and accessibility barriers that can be impossible to overcome without a personal vehicle or pet-friendly public transit. Recruitment strategies also play a role, with some researchers leveraging social media platforms popular with younger generations, such as Instagram or TikTok, while others connecting with older generations through dog interest groups on

Facebook or via referrals within local training schools or breed clubs. Even within these communities, sociodemographics and individual beliefs about animal minds may drive self-selection for research participation. Guardians who are more interested in dog behavior, value novel enrichment opportunities, share a closer bond with their dog, or have invested more in training, *while also* having the disposable time and resources, will be disproportionately represented in research samples. Additionally, certain samples may favor some breeds, as demographics and cultural trends can shape preferences for certain physical features or behavioural traits in companion dogs (e.g., Özcan et al., 2017). On the other hand, while well-socialized, confident dogs with highly invested guardians may be overrepresented, guardians with socially anxious dogs that find transportation stressful, will be underrepresented. This suggests that researchers should collect and report standard demographic data about the dogs and owners that they study, as well as information about recruitment strategies. These data can then be compared across locations to account for potential site differences as well as to the local population demographics to assess local sample representativeness.

Because of these factors, only a narrow range of companion dogs and guardians are included in research at a given site. These constraints are similar to biases observed in human psychological research, where findings are often based on readily available student samples rather than representative cross-sections of the population (Hanel & Vione, 2016). Therefore, the current state of knowledge in canine science does not explain dogs as a species and limits our understanding of human-animal interactions. These WEIRD biases in guardian participation are further compounded by their companion dogs fitting several of the categories described in the STRANGE framework (Webster & Rutz, 2020). Particularly relevant categories in the context of human-animal interaction research include *social background, rearing history, genetic make-up,* and *experience.* Most dogs included in today's canine research could be described as *strange* in these categories, and this limited representativeness poses another threat to reproducibility.

Research subjects are frequently from recognizable single breeds or intentional hybrids originating in Europe or North America (e.g., Border Collie or Goldendoodle). These dogs have experienced selective breeding and genetic bottlenecks (Marsden et al., 2015; Dutrow et al., 2022) causing significant, breed-level differences in behavior and personality (MacLean et al., 2019). Further, companion animals' interactions with their environment are a reflection of their human guardian's preferences, socioeconomic status, and lifestyle. In contrast to this *strange* subpopulation, the majority of domestic dogs in the world are free-living, freely-breeding individuals whose patterns of behavior are determined by their need to access food, shelter and mating opportunities (e.g., Sen Majumder et al., 2014). While free-living dogs make up ~75% of the world's domestic dog population (Hughs & Macdonald, 2013), they are rarely studied outside of behavioral ecology and conservation contexts. As a result, we know little about normative dog behavior and cognition in the general population.

The final dimension limiting representativeness in canine science comes from the researchers themselves. Often from WEIRD cultures or trained in Western scientific traditions, canine scientists may find themselves perpetuating *strangeness* in their study populations because of their educational background or current research environment. Further, a researcher's access to institutional resources as well as their personal connection to dogs may shape their choice of study population and methods, such as working in shelters, visiting guardian homes, or recruiting for in-lab participation. At a higher level, cultural values and perspectives also drive research priorities, define incentive structures, and

methodological innovation, which do not represent the breadth of epistemologies on animal behavior and human-animal interactions. For example, a country's emphasis on security may generate funds to study detection dogs, which focuses effort to work on odor detection. To broaden the field, corresponding shifts need to happen in the funding landscape and general scientific community to create space for globally relevant and representative research.

Big Team Science and Other Solutions

Big team science has the potential to overcome many challenges of generalizability and representativeness within canine science. Big team science is large-scale collaborative research conducted across multiple independent research sites. In comparative psychology, big team science initiatives such as ManyDogs, ManyBirds, and ManyPrimates actively work to reduce sampling biases, increase transparency, and integrate diverse research perspectives to advance their fields (ManyPrimates, 2019; Lambert et al., 2022). In the canine science context, leveraging the power of larger samples and increasing the diversity of sampled populations and researcher identities can address many of the issues identified above, in addition to the systemic issues with justice, equity, diversity, and inclusion in research.

Big team science provides a solution to low generalizability across research sites. Instead of sites conducting studies independently and consistently reinforcing their own results, big team science allows us to combine data and collectively investigate widely applicable effects. In addition to testing for generalizability, large-scale collaboration allows us to observe and test what features across sites might account for differences we observe. Thus, facilitating both the broader assessment of general phenomenon and the smaller-scale investigation into what factors may moderate observed effects.

Large-scale collaborations can address fundamental limitations in participant representativeness. Initiatives such as ManyDogs encourage researchers to collectively develop best practices for recruitment, integrate new communication modalities and outreach tools, and innovate methods specifically designed to improve sample diversity for both guardians and dogs across a range of cultures and research contexts. For example, to complement in-lab studies, researchers could also test a subset of dogs in the home, or via online participation. These approaches reduce transportation costs and remove some accessibility barriers while increasing the possibility for anxiety-prone or fearful dogs to participate. Online methods may be particularly effective at increasing the representation of dogs and guardians, however they necessarily limit the type of research that can be done. Notably, big team science cannot solve all representativeness issues as guardian self-selection biases may persist, with participation skewed towards highly motivated individuals with over-trained dogs.

The international nature of big team science facilitates co-creation of globally relevant research questions, mitigates *strange* factors in certain populations, and promotes researcher diversity and inclusion. Large-scale collaborative research builds up partnerships spanning the Global South and North, simultaneously improving scientific rigor, and researcher equity and inclusion. In the special case of domestic dogs, we propose that multi-site collaborative studies may be the key for understanding the species as a whole. Multi-site studies open the door to compare differences in the ethology of free-living dogs from different populations, diverse cultures, and habitats. Including dogs from different regions, such as Northern regions of Canada, street dogs in Mexico or Italy, or village dogs in Morocco will advance canine science by directly addressing some of the *strange* problems with companion dog

populations. Big team science creates unparalleled opportunities to address some of the biases in canine science (e.g., Winder et al., 2025), increasing research reproducibility and representativeness.

Though big team science offers exciting possibilities, it comes with a number of challenges (Forscher et al., 2022). Research funding is scarce for large, distributed initiatives, and research groups are generally expected to use their existing resources (i.e., space, supplies, personnel) in multi-lab collaborations. Between the Global South and North there are marked resource inequities, which imposes additional barriers to inclusion. Additionally, big team science projects are necessarily slower moving and do not align with traditional incentive structures in academia, making it particularly risky for early career researchers or those facing high publication demands and low job security to invest time without quicker outcomes to advance in their careers.

Final Considerations

Canine science has severe limitations and biases, both in the researchers carrying out the work and in the populations being studied. In particular, generalizability is hampered by limited access to diverse samples and representativeness of the research participants is influenced by socioecological factors and biological traits. Current findings are difficult to generalize even within the studied populations (primarily dogs the Global North), due to breed differences and cultural variation in guardianship and training norms. Further, the limited inclusion of guardians from diverse backgrounds and socioeconomic classes and narrow focus on companion animals prevents a deep understanding of domestic dogs as a species.

Big team science initiatives offer the potential for expanding collaborations and overcoming systematic issues with researcher and population diversity and inclusion. While there is still much that can be improved, big team science is starting to change the norms around research collaborations and open the door for underrepresented research populations and researchers to have a voice.

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